

Haoran Ding

Chesapeake, VA | hding192@gmail.com | (+1) 585-309-3306 | [Google Scholar](#)

RESEARCH PROFILE

Machine learning researcher specializing in large language models, graph-based learning, attention mechanisms, and interpretable AI systems. Research focuses on integrating global structural information with transformer architectures, improving model interpretability, and designing scalable algorithms for long-document understanding and molecular representation learning. Extensive publication record, including methodological contributions in NLP, AI/ML, and interdisciplinary collaborations in biomedical informatics. Experienced in leading model design, algorithm development, and reproducible research pipelines.

RESEARCH INTERESTS

- Large Language Models and Hybrid LLM Architectures
- Graph-Augmented Neural Methods
- Attention Mechanisms and Representation Learning
- Interpretable and Trustworthy AI
- Multi-Agent and Global Information Modeling
- AI for Scientific Discovery

EDUCATION

Ph.D., Computer Engineering - Purdue University, 2024

Advisor: Xiao Luo, Ph.D.

Dissertation: LLMs for Unsupervised Keyphrase Extraction and Biomedical Data Analytics

M.A., Mathematics (Statistics) - Binghamton University, 2019

B.S., Mathematics - Nazareth University, 2015

RESEARCH EXPERIENCE

Postdoctoral Researcher

Binghamton University | 2025.3 – 2025.8

PI: Nancy Lan Guo, Ph.D.

- Led the design and implementation of an interpretability framework integrating attention and SHAP analyses for molecular property prediction models.
- Engineered custom extensions to the fast-transformers library to enable layer-specific attention extraction from large-scale SMILES-based models, preserving full compatibility with existing inference workflows.
- Developed a dataset-level token importance aggregation algorithm that corrects for prediction confidence, token frequency, and sequence-length bias, producing reliable feature rankings across millions of molecular samples.
- Adapted SHAP (SHapley Additive Explanations) for both Transformer- and graph-based molecular encoders (SMI-TED, SAMPN, and MAMMAL), introducing multi-run averaging and sampling strategies to stabilize attributions.

- Designed structure-aware functional group-level attribution by mapping token SHAP values to RDKit-derived SMARTS patterns, yielding chemically interpretable group-level insights.
- Served as primary developer for the attention/SHAP ranking modules used in current publications, providing analytical methodology, implementation, and codebase support.
- Provided statistical consultation on study design, data validation, and reproducibility across multiple biomedical projects.
- Mentored PhD and undergraduate researchers on analytical design, programming practices, and LLM-based clinical data extraction.

Research Assistant

Purdue University | 2019.7–2024.6

- Developed graph-augmented and attention-based ranking models for unsupervised keyphrase extraction (AttentionRank, AGRank, LongDocRank), improving F1 performance by up to 11% over prior baselines on benchmark datasets.
- Designed hybrid LLM + graph frameworks to mitigate long-context limitations in transformer models, integrating global structural signals to enhance robustness on long-document corpora.
- Built hierarchical attention-based deep learning models for structured prediction tasks, achieving 98–99% classification accuracy in applied biomedical settings.
- Integrated structured EHR variables with unstructured clinical notes using NLP-based feature fusion methods, improving outcome prediction AUC by 3–4% over structured-only approaches.
- Conducted systematic model evaluation including ablation studies, cross-validation, and statistical validation to ensure robustness and reproducibility.
- Implemented scalable data preprocessing and experiment management pipelines using Python, SQL, and AWS S3 to support large-scale modeling workflows.

PUBLICATIONS (AI / ML)

- **Ding, Haoran**, and Xiao Luo. "LongDocRank: graph-augmented large language models for unsupervised keyphrase extraction from long documents." *Journal of Big Data* (2026).
- **Ding, Haoran**, and Xiao Luo. "AGRank: Augmented Graph-based Unsupervised Keyphrase Extraction." *Proceedings of the 2nd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 12th International Joint Conference on Natural Language Processing*. 2022.
- **Ding, Haoran**, and Xiao Luo. "AttentionRank: unsupervised keyphrase extraction using self and cross attentions." *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*. 2021.
- Jing, Taotao, Haifeng Xia, Renran Tian, **Haoran Ding**, Xiao Luo, Joshua Domeyer, Rini Sherony, and Zhengming Ding. "InAction: Interpretable action decision making for autonomous driving." *Computer Vision—ECCV 2022: 17th European Conference, Tel Aviv, Israel, October 23–27, 2022, Proceedings, Part XXXVIII*. Cham: Springer Nature Switzerland, 2022.

- **Ding, Haoran**, and Xiao Luo. "Attention-based unsupervised keyphrase extraction and phrase graph for covid-19 medical literature retrieval." *ACM Transactions on Computing for Healthcare (HEALTH)* 3.1 (2021): 1-16.

INTERDISCIPLINARY APPLICATIONS

- Altherr, Cody A., Xiao Luo, **Haoran Ding**, Ziyue Liu, Stuart J. Warden, Ranjani N. Moorthi, and Erik A. Imel. "Using an electronic frailty index and patient reported outcomes to predict sarcopenia risk." *Scientific Reports* 15, no. 1 (2025): 35372.
- Luo, Xiao, **Haoran Ding**, Stuart J. Warden, Ranjani N. Moorthi, and Erik A. Imel. "Integrate data-driven and knowledge-driven approaches to analyze clinical notes with structured data for sarcopenia detection." *Health Informatics Journal* 30, no. 4 (2024): 14604582241300025.
- Luo, Xiao, **Haoran Ding**, Andrea Broyles, Stuart J Warden, Ranjani N Moorthi, Erik A Imel. "Using machine learning to detect sarcopenia from electronic health records." *Digital Health* 9 (2023): 20552076231197098.
- Luo, Xiao, Lena Ara, **Haoran Ding**, David Rollins, Raghu Motaganahalli, and Alan P. Sawchuk. "Computational methods to automate the initial interpretation of lower extremity arterial Doppler and duplex carotid ultrasound studies." *Journal of Vascular Surgery* 74, no. 3 (2021): 988-996.
- Luo, Xiao, **Haoran Ding**, Matthew Tang, Priyanka Gandhi, Zhan Zhang, and Zhe He. "Attention mechanism with bert for content annotation and categorization of pregnancy-related questions on a community Q&A site." *2020 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*. IEEE, 2020.
- Lu, Yu, Xiao Luo, Zhan Zhang, **Haoran Ding**, and Zhe He. "Retrieving Lab Test Related Questions from Social Q&A Sites by Combining Shallow Features and Deep Representations." *AMIA Annual Symposium Proceedings*. Vol. 2020. American Medical Informatics Association, 2020.

TECHNICAL EXPERTISE

- **Programming:** Python, PyTorch, TensorFlow, R, SQL
- **Modeling:** Transformers, Attention Mechanisms, Graph Learning, SHAP
- **Data Systems:** AWS, GCP, Snowflake, SSIS, Power BI
- **Statistical Methods:** Regression, Survival Analysis, Experimental Design